

PUNIT GUPTA

PRINCIPAL SOFTWARE ENGINEER

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SUMMARY

iOS Engineer with nearly 10 years of experience architecting high-impact, scalable mobile solutions across fintech, banking, aviation, digital currency, and e-commerce. Deep expertise in iOS system design, mobile architecture (MVVM, VIPER, Protocol-Oriented Programming, Clean Architecture), and performance optimization. Sets technical direction and owns mobile architecture end-to-end—from translating ambiguous product requirements into executable designs through implementation, integration, release, and production support. Architected platforms spanning multi-currency payments, biometric authentication, real-time connectivity (BLE, Web Sockets), secure offline-first workflows, and blockchain-enabled digital asset transactions. Brings practical experience in AI and LLM-assisted development, including Claude Code, ChatGPT, and MCP, to accelerate feature delivery, code analysis, and documentation.

CORE SKILLS

iOS Architecture and Design: SOLID Principles, Modular Architecture, Design Systems, Mobile Architecture, MVC, MVVM & VIPER, Apple Connectivity, Apple Developer Console, Reusable Frameworks with Xcode.

Mobile Platform Engineering: Swift 3/4/5/6, Objective-C, Production SwiftUI, UIKit, Cocoa Touch, Auto Layout, Swift Package Manager, CocoaPods, State Driven Navigation, Deep Linking, Universal Links, Core Graphics, Interface Builder.

Concurrency and Performance: Concurrency (Async/Await), Combine, GCD, Threading Models, Memory Optimization, Instruments Profiling, Performance Tuning, App Startup Time Optimization, Application Stability, Frozen Frames Prevention.

Networking and Integration: URLSession, REST APIs, Offline-first, Web Services Integration, SOAP, WebSocket, GraphQL, Socket Programming, API Design, Caching, Retry Mechanisms, Secure Data Exchange.

FinTech and Payments: Apple Pay, Stripe, Plaid, Coinbase, WalletConnect, UPI, PayPal, Credit Card Processing, In-App Purchase, 3D Secure and Digital Assets.

Analytics and Observability: Analytics (Firebase, Adobe), Crash Analysis & Monitoring, Performance Monitoring, Event Tracking, Operational Insights.

Testing and Quality Engineering: Unit Testing, Snapshot Testing, Integration Testing, TDD, XCTest, Test Coverage, Debugging, Code Quality, Swift Lint, Technical Debt Reduction, Release Readiness and Wireshark.

Cloud, CI/CD and DevOps: CI/CD Pipelines, Git, Build Automation, Release Management, Feature Flags, Remote Configuration, Continuous Delivery, Provisional Profiles, App Store Deployment, App Store Connect.

Product Domains: FinTech, Banking, Digital Payments, Web3/Crypto, Travel and Booking Platforms, Donation Platforms, Enterprise Applications, TestFlight.

Mobile Engineering Leadership: Technical Leadership, Problem Solving, Decision Making, Architecture Reviews, Code Reviews, Mentorship, Agile/Scrum, Stakeholder Management, Cross Functional Collaboration, Product & Feature Delivery, Management (Team, Project, Risk, Budget & Risk), SDK architecture, Conflict Resolution, Product Strategy, Cost Optimization.

AI and Developer Productivity: OpenAI Codex, ChatGPT, Claude, Cursor AI, GitHub Copilot, Gemini, AI/ML assisted Development, Code Generation, Refactoring, Unit Test Generation, Documentation, Debugging.

BLE communication: BLE Layer, device pairing, authentication, synchronization, OTA updates, and real time data streaming, device controls, and sensor data access, design wellness features such as heart rate, SpO₂, sleep, activity, GATT, Wearable, HealthKit, nRF Connect and stress monitoring.

Security and Authentication: Data encryption and mobile security, including AES-256, RSA, ECC, digital signatures, tokenization, TLS/HTTPS, SSL pinning, mTLS, Perfect Forward Secrecy (PFS), data at rest/in transit protection, JWT, OAuth 2.0, bearer/access/refresh tokens, Face ID, Touch ID, Web Auth, MITM attack prevention, public/private and secret key management, jailbreak detection, FIDO, PCI DSS, SOC 2, Apple Keychain, Secure Enclave, Passkeys, secure storage, authentication flows, OTP verification, and payment security.

EXPERIENCE

Staff Software Engineer -> Principal Software Engineer | BeneFi

July 2024 - Present

- Architected extensible fiat, stock, and crypto payment solutions, compliance, scalability, and future platform extensibility.
- Built blockchain-enabled self-custody wallets and digital asset transaction infrastructure, integrated Stripe, Plaid, Coinbase, and WalletConnect for secure FinTech capabilities.
- Introduced Clean Architecture, and Protocol-Oriented Programming to establish engineering standards; architected end-

to-end user journeys including authentication, onboarding, and state-driven navigation.

- Implemented token lifecycle management, feature flags, remote configuration, and observability across the application; implemented analytics, observability, and crash monitoring to improve platform reliability and engineering excellence.
- Applied performance optimization and reusable engineering practices to ensure stability; delivered data-driven mobile experiences that accelerated feature delivery, improved application quality, and enhanced user engagement.
- Optimized application performance using Instruments (Memory, CPU, and I/O) and mentored engineers on best practices; led architecture reviews, technical design discussions, and code reviews across the engineering organization.
- Utilized Claude Code for intelligent code analysis, security pattern detection, and architecture optimization across payment and blockchain modules; collaborated with Product, Design, Backend, QA, and leadership teams to drive technical strategy and successful product delivery.

Software Development Engineer III - Mobile | Cleartrip (Flipkart)

Feb 2024 – Jun 2024

Key Achievement: First launch of Cleartrip Arabia.

- Led iOS architecture, system design, and technical roadmap for Cleartrip Middle East and Flyin applications; drove scalable, modular, and production-ready solutions while partnering with engineering leadership on technical strategy and release planning.
- Reduced application crashes by 15% through systematic analysis and memory management; optimized API integrations and refactored legacy modules, significantly reducing technical debt and strengthening release confidence.
- Leveraged AI for debugging, performance optimization, rapid prototyping, technical documentation, and architecture decision records; architected and delivered mission-critical travel capabilities including booking flows and GraphQL integration.
- Built secure payment workflows with Apple Pay, Credit Cards, 3D Secure, encryption, and secure storage; implemented analytics, operational reporting, and baggage & passenger insurance features, improving user engagement, conversion, and transaction reliability.
- Owned technical delivery from architecture through production, leading architecture reviews, code reviews, and mentoring engineers; collaborated cross-functionally with Product, Backend, QA, Design, and business stakeholders to ensure production readiness and successful market launches.
- Designed reusable platform components and internal frameworks, improving development velocity, application consistency, and release quality; enabled faster feature delivery across both high-traffic travel applications through reusable components and standardized practices.

Sr. iOS Consultant | Emirates National Bank (Via Spare IT Dubai)

Aug 2023 – Feb 2024

Key Achievement: Design Whole app analytics and Notification (Insider SDK) for marketing campaign.

- Developed scalable iOS banking modules for the ENBDX platform, Cocoa Touch, and reusable private frameworks, improving code quality and maintainability
- Integrated Insider SDK for rich push notifications, delivering personalized transaction updates, feature announcements, and customer engagement campaigns.
- Implemented Adobe Analytics across the core platform to track user journeys, funnel drop offs, feature adoption, conversion metrics, and product usage insights.
- Designed reusable private libraries for networking and shared business logic while implementing multilingual support with English and Arabic (RTL) localization.
- Enhanced the banking experience through Core Animation, interactive UI, performance optimization, and continuous feature enhancements for a stable production application.

Senior. iOS Developer | Paytm Payments Bank

Jan 2022 – Aug 2023

- Designed and delivered the Add Money module, improving payment workflows and transaction reliability; engineered secure Wallet features, enhancing digital payment and account management experiences.
- Architected and implemented Rupay Credit Card UPI onboarding and payment integration with the UPI ecosystem; developed Aadhaar OTP based UPI MPIN setup, ensuring secure authentication and regulatory compliance.
- Applied Unit Testing and clean code practices to improve application quality and maintainability.
- Leveraged the Backbase Mobile SDK to build scalable Banking applications, enterprise grade digital banking solutions.

Senior iOS Developer | Birlasoft pvt ltd

Feb 2021 – Jan 2022

- Developed and enhanced iOS modules, Cocoa Touch, and Objective C, including refactoring private Cocoa Pods to improve maintainability and code quality; optimized application performance through architecture improvements and code refactoring, reducing technical debt while improving stability and responsiveness.
- Designed and implemented an offline peer-to-peer synchronization system where crew devices connected to an onboard Wi-Fi router to share itinerary, order, and payment updates in real time without internet connectivity; improved cabin

crew efficiency by ~20% through automated in-flight ordering, payment processing, and synchronized offline data sharing across connected devices.

- Built key airline features including meal collection, voucher scanning, order correction workflows, and MileagePlus payments with seamless online and offline support for flight attendants; strengthened operational efficiency by enabling crew to manage orders, corrections, and payments in real time.
- Collaborated closely with Product, Backend, and QA teams to deliver reliable end-to-end features and ensure production readiness; coordinated cross-functional requirements, resolved blockers, and contributed to successful market launches.

iOS Developer | Aeonic Edge pvt ltd

Jun 2019 – Jan 2021

- Designed and developed iOS applications from scratch (Dmystifi and School ERP), Cocoa Touch, Auto Layout, SQLite, and Objective C maintenance; optimized local data persistence using SQLite for offline access, synchronization, and improved application responsiveness.
- Built the Dmystifi online learning platform with live and recorded classes, AVFoundation, TLS, playback, learning progress tracking, and SQLite based offline content storage; implemented student learning validation to ensure lessons were completed sequentially, preventing users from skipping ahead while tracking total watch time and course completion.
- Developed a real time school bus tracking system using Core Location, Apple Maps, and custom route animations, enabling parents to monitor bus locations, receive ETA updates, and get push notifications when the bus approached their pickup stop or reached its destination, improving child safety.
- Integrated multiple payment gateways, Apple QR code, interactive charts, and custom animations to enhance user experience and application functionality.

iOS Developer | Prime lite technology pvt Ltd

May 2017 – May 2019

- Designed and developed iOS applications from scratch (Dmystifi and School ERP), Cocoa Touch, Auto Layout, SQLite, and Objective C maintenance; optimized local data persistence using SQLite for offline access, synchronization, and improved application responsiveness.
- Built the Dmystifi online learning platform with live and recorded classes, video playback, learning progress tracking, and SQLite based offline content storage; implemented student learning validation to ensure lessons were completed sequentially, preventing users from skipping ahead while tracking total watch time and course completion.
- Developed a real time school bus tracking system using Core Location, Apple Maps, and custom route animations, enabling parents to monitor bus locations, receive ETA updates, and get push notifications when the bus approached their pickup stop or reached its destination, improving child safety.
- Integrated multiple payment gateways, Apple QR code, interactive charts, and custom animations to enhance user experience and application functionality.

EXTERNAL PROJECT

TRADEDOG | India

May 2023

- Designed and developed a research-focused iOS application using Storyboard and Auto Layout to provide future investment analysis, helping users identify which digital currencies are best suited for long-term growth.
- Integrated REST API calling to fetch, display, and manage research posts on digital currency, enabling users to access institutional-grade insights and market data in real time.
- Implemented Socket programming for live price updates and real-time chat, ensuring users receive current market trends and can interact with the community while analyzing investment opportunities.
- Built graphical representations with live market data using custom diagrams and interactive charts, empowering investors to visualize trends and make data-driven decisions about future digital asset investments.

SPARSA | USA

Sep 2020

- Developed and maintained a medical healthcare iOS application using Objective-C, collaborating with cross-functional teams to fix critical bugs, improve stability, and ensure reliable device-to-app communication across multiple iOS versions.
- Integrated IoT wearable devices to monitor user health metrics—including pulse rate and blood pressure—via **Bluetooth/BLE**, enabling real-time data syncing and proactive health management for end users.
- Designed and implemented comprehensive health record management, allowing users to track daily, weekly, and monthly health trends, with secure local persistence and historical data retrieval for personalized insights.
- Built a motivational UI dashboard using interactive graphs and custom emoji rewards, visualizing walking steps and health milestones to encourage users to achieve their daily physical activity goals.

EDUCATION

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| • Master of Computer Applications (MCA) , GGSIPU University | 2014 – 2017 |
| • Master of Business Administration (MBA-IT) , SMU | 2018 - 2020 |

LEADERSHIP SKILL

Strategic Planning, Cross-Functional Leadership, Secure Coding Practices, Test Automation, Revenue Growth, Strategic Thinking, Prompt Engineering, Quality Assurance, Performance Testing, Regression Testing, Data Privacy, Event-Driven Architecture, App Launch Time, App Size Optimization, Battery Optimization, Network Performance, Network conditioner, Pre-caching, Lazy Loading, Background Tasks, Large Scale System Design, On-Call Rotation, Advanced Debugging tool, Latency Optimization, Pub-Sub/Streaming Protocols, Protocol Buffers, Modern Build Systems, Top-Chart App Store Pedigree, Designing GenAI Capabilities, Agentic Workflows, Cross-Platform Awareness, Accessibility, Fastlane, open source contribution.

External team support: Android, Kotlin, Java, Node JS, QA, Business Team, Azure, Oracle Cloud, AWS Amplify, Backend less, Docker, Kubernetes, Fastlane, Launch Darkly, Mix panel, Amplitude, Sentry, New Relic, Insta bug, .NET MAUI, Razor pay, Digital health, connected healthcare, remote patient monitoring, medical devices, application lifecycle management, React Native, Safe, Less, device trust, SCA, experimentation frameworks, regulator/audit exposure, Computer Vision/CV, PII, integration testing, Ruby, Fastlane, open-source contribution, React Native, Flutter, Cordova, A/B testing, Kotlin Multiplatform/KMP, XCFramework, semantic versioning, Circle CLI, Spring Boot, HTML5, XML, SQL, Docker, Kubernetes, Kafka, Microsoft Office.

CERTIFICATE

- **Java Certified**, Ducat Noida
- 2014